

Lab 1: Requirements Engineering & UML Use-Case Modelling

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Problem statement 16: Remote Patient Vitals Alert & Monitoring App

The application monitors post-operative patients using SpO2, heart rate and blood pressure readings. It detects configured threshold breaches and alerts the on-call caregiver, escalating unanswered alerts through a caregiver matrix.

1. Functional requirements

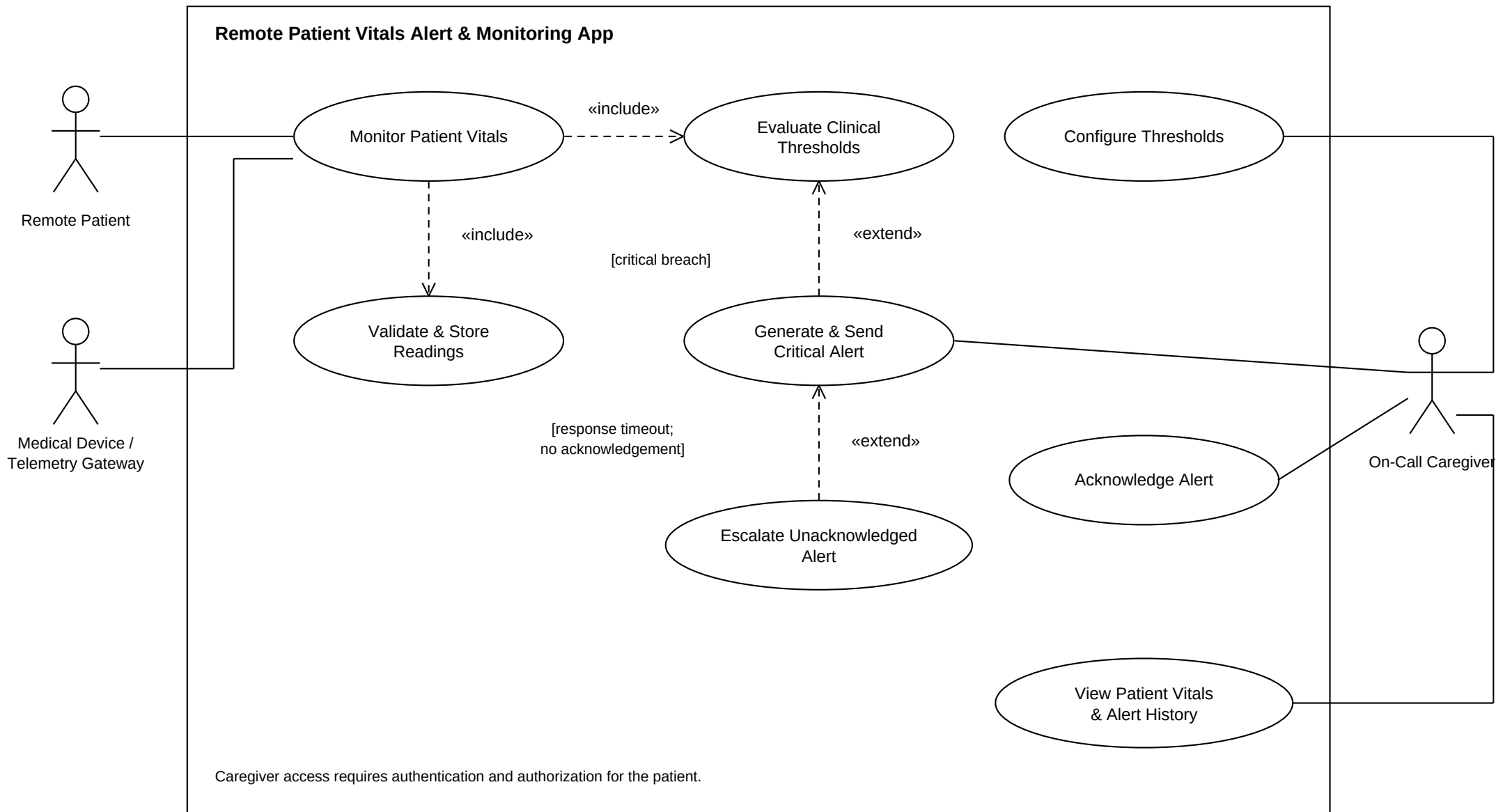
ID	Type	Description	Priority	Acceptance criteria	Rationale
FR-001	Functional	The system shall receive, validate and store patient SpO2, heart rate and blood pressure readings with patient identity and measurement timestamps.	High	Pass: Valid readings are stored against the correct patient with timestamps. Invalid or incomplete readings are rejected and logged. Fail: A valid reading is lost, assigned to the wrong patient or stored without a timestamp.	Provides reliable input for continuous monitoring.
FR-002	Functional	The system shall allow authorized caregivers to configure patient-specific thresholds and flag critical breaches within 2 seconds of receiving a valid reading.	High	Pass: With a heart-rate upper limit of 140 BPM, a 145 BPM reading is flagged within 2 seconds. Readings within all configured limits are not flagged. Fail: A breach is missed or detection exceeds 2 seconds.	Supports timely detection using patient-specific limits.
FR-003	Functional	The system shall generate a critical alert and notify the assigned on-call caregiver when a critical threshold breach is detected.	High	Pass: The assigned caregiver receives the patient ID, affected vital, measured value, breached threshold and detection time. Fail: An alert is absent, misdirected or incomplete. Delivery timing is assessed under NFR-002.	Supplies the details needed for a caregiver response.

Requirements Table (continued)

ID	Type	Description	Priority	Acceptance criteria	Rationale
FR-004	Functional	The system shall escalate unanswered alerts through the caregiver matrix after each configured timeout and stop further escalation after acknowledgement.	High	Pass: With a 60-second test timeout, the next caregiver is notified when the timeout expires. Each escalation is logged. Acknowledgement cancels pending escalation. Matrix exhaustion is logged and the alert stays unacknowledged. Fail: Escalation is skipped, out of order or continues after acknowledgement.	Provides a response path when a caregiver is unavailable.
FR-005	Functional	The system shall provide an authenticated caregiver dashboard to view assigned patients' latest vitals, review alert history and acknowledge active alerts.	High	Pass: Authorized caregivers can access assigned patients and acknowledge alerts. The system records caregiver identity and time and sets status to Acknowledged. Access to unassigned patients is denied. Fail: Unauthorized data is exposed or acknowledgement is recorded incorrectly.	Supports secure monitoring and records responsibility.
NFR-001	Performance & availability	The ingestion gateway shall support at least 500 concurrent telemetry streams and achieve 99.99% availability per calendar month.	High	Pass: A 60-minute test with 500 streams, each sending one reading per second, loses no valid readings and meets FR-002. Monthly monitoring shows availability of at least 99.99%. Fail: Either target is missed. Availability = available service time / total calendar-month time.	Supports the required patient load with minimal interruption.
NFR-002	Reliability & auditability	At least 95% of critical notifications shall be delivered within 5 seconds of anomaly detection. Alert generation, delivery outcomes, escalation and acknowledgement shall be audited.	High	Pass: Under the 500-stream test load, at least 950 of 1,000 critical notifications have confirmed delivery within 5 seconds. Every tested event records alert ID, event type and timestamp; acknowledgements also record caregiver identity. Fail: Delivery is below 95% or required audit events are missing.	Makes notification reliability measurable and supports incident review.

Test assumptions: The 60-second timeout, one-reading-per-second stream rate and test duration are proposed verification settings. Clinical thresholds and operational escalation timeouts are configurable. Acknowledged means a caregiver has accepted the alert; it does not mean the patient's condition is resolved.

2. UML Use-Case Diagram



3. Use-Case Flow Specification

UC-01: Generate & Send Critical Alert

Primary actor: On-Call Caregiver

Other participants: Remote Patient (monitored subject); Medical Device / Telemetry Gateway (telemetry source).

Trigger: A patient vital reading arrives for threshold evaluation.

Related requirements: FR-001 to FR-005; NFR-002.

Preconditions

1. The patient is registered and linked to an authenticated monitoring device.
2. Patient-specific thresholds, an ordered caregiver matrix and response timeouts are configured.
3. The monitoring and notification services are operational. The caregiver must authenticate and be authorized before viewing patient details or acknowledging an alert.

Main success scenario

1. The device transmits the patient's vital reading and measurement timestamp.
2. The system validates the reading, associates it with the patient and stores it.
3. The system compares the reading with the configured thresholds and flags the breach within 2 seconds of receipt.
4. The system creates an alert with the patient ID, vital value, breached threshold and detection time. Its status is Pending Acknowledgement.
5. The system notifies the first caregiver in the matrix, records the delivery outcome and starts the configured response timer. Delivery is measured against NFR-002.
6. The caregiver opens the authenticated dashboard, reviews the alert and acknowledges it before the timeout.
7. The system records the caregiver's identity and acknowledgement time, changes the status to Acknowledged and cancels pending escalation.
8. The updated alert and audit events appear in the patient's alert history.

Alternate flow A1: Caregiver does not acknowledge

A1.1 At step 6, the configured response timeout expires without acknowledgement.

A1.2 The system records the timeout and escalates the alert to the next caregiver in the matrix. The alert remains Pending Acknowledgement.

A1.3 The system records delivery and starts the next response timer. If the caregiver acknowledges, resume at step 7; otherwise repeat this alternate flow.

A1.4 If no caregiver remains, the system records matrix exhaustion and keeps the alert visible as unacknowledged for follow-up.

Postconditions

Success: The alert is Acknowledged, caregiver identity and time are recorded, and pending escalation is stopped. Clinical resolution is not implied.

Alternate outcome: If the matrix is exhausted, the alert remains unacknowledged. Every attempted notification and escalation is retained in the audit history.